

NAGAKANNI MUTHUVENUGOPAL

Ph.D. in Biotechnology (pursuing)

Alagappa University

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ACADEMIC PROFILE - DEGREE PROGRAMS

Year	Degree	Specialization	Institution
2020	M.Sc.	Biotechnology	Alagappa University, Karaikudi, Tamil Nadu, India
2018	B.Sc.	Biochemistry	The American College, Madurai, Tamil Nadu, India

EDUCATION - TESTS AND CERTIFICATE PROGRAMS

2020	Graduate Aptitude Test (GATE) in Life Sciences
2019	SWAYAM online certification course - Biomolecules: Structure, Function in Health and Disease

RESEARCH EXPERIENCE:

2021- Till date	Ph.D. Scholar Mentor: Prof. K. Pandima Devi, Alagappa University, India. Dissertation work: Screening and evaluation of phytocompounds targeted against lung adenocarcinoma: <i>in silico</i> and pre-clinical evaluation.
2019-21	Graduate Student Mentor: Dr. Manisha Banerjee, Bhabha Atomic Research Centre, India. M.Sc. thesis work: Protein and Gene Expression Analysis of Cyanobacterium Anabaena PCC7120 Under Different Stress Conditions.
2019	Graduate Student Mentor: Dr. Suresh Govindhan, N. Rama Varier Ayurveda Foundation, AVN Ayurveda Formulations Pvt Ltd., India.

	Assessment and Identification of Bioactive Compounds from Medicinal plants using HPTLC analysis, medicine to tablet printing, its quality control and validation to use it as a potent drug.
2017-18	Undergraduate Student Mentor: A. K. V. Sama, Department of Biochemistry, The American College, India. Analysis of physicochemical properties of seaweed Sargassum sp. and the usage of the seaweed extract as a potential biofertilizer in enhancing the growth of plants.
2017	Undergraduate Student Apollo Lab Services, Apollo Hospitals, Madurai, India. Intern: Analysis of various patient samples in the department of Clinical Microbiology and Immunology, Clinical Biochemistry, Haematology, Histopathology.

PUBLICATIONS

PEER REVIEWED ARTICLES:

1.	2025	Jafni, S., Nagakanni, M. , Indhirakumar, B., Rani, R. K. S., Preethi, S., Hemamalini, V., Vanathi, G., Saraswathy, S. D., & Devi, K. P. (2025). Synergistic neuroprotective effects of Vitexin and Thymol against Okadaic acid-induced neurotoxicity: Computational and In vitro evaluation. <i>Molecular biology reports</i> , 53(1), 43. https://doi.org/10.1007/s11033-025-11190-y
2.	2025	Nagakanni Muthuvenugopal , Sangita Behera, Soundarya Rani Rajendra Kumar, Indhirakumar Balakrishnan, Pandima Devi Kasi. Farnesol induces apoptosis, LC3B/SQSTM1 mediated regulation of autophagy and downregulates anaerobic glycolysis through suppression of LDH and PKM in A549 lung adenocarcinoma cells. <i>Medical Oncology (Accepted)</i>

REVIEW ARTICLES:

1.	2025	Roles of Nrf2-p62 Pathway in Cancer Drug Resistance: Insights into Mechanisms and Therapeutic Strategies. (Under Communication)
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INTERNATIONAL LEVEL PRESENTATIONS:

23rd - 25th Jan, 2023	International Conference on Innovations and Applications in Biomedical and Health Sciences - conducted by PD and Research Department of Biotechnology and Bioinformatics, Holy Cross College, Tiruchirapalli, India in collaboration with Department of Anatomy, Yong Lin School of Medicine, National University of Singapore and Department of Endocrinology, Dr.ALM PG IBMS, University of Madras, Taramani Campus, Chennai. ● Nagakanni Muthuvenugopal, Pandima Devi Kasi Unveiling the Differentially Expressed Genes and Therapeutics using <i>In silico</i> Approaches Against Early Stage Non-Small Cell Lung Cancer.
6th & 7th February, 2025	2nd International Conference on Recent Trends in Vaccines and Biomaterials for Animal Health (RTVBAH-2025) - conducted by the Department of Animal Health and Management, Alagappa University, Karaikudi. ● Nagakanni Muthuvenugopal, Jafni Sakthivel, Indhirakumar Balakrishnan, Soundarya Rani Rajendran, Pandima Devi Kasi* Unveiling the Therapeutic Effect of Essential Oil (A) Through <i>in vitro</i> LC- MS/MS analysis Against non-small cell Lung Cancer. ● <u>S. Jafni</u>, M. Nagakanni , B. Indhirakumar, R. Soundharya Rani and K. Pandima Devi. Evaluation of Hesperidin Methyl Chalcone as a potential drug candidate against Alzheimer's Disease ● <u>Indhirakumar Balakrishnan</u>, Jafni Sakthivel, Nagakanni Muthuvenugopal, Soundarya Rajendran, Pandima Devi Kasi. Molecular Docking and Pharmacokinetic Analysis of Phytocompounds Targeting tau Hyperphosphorylation in Alzheimer's Disease
28th February and 1st March, 2025	“6th International Conference on Genome Biology: Cell - Organism - Environment & 56th Aqua-Terr Annual Day Celebration (ICGB6_AquaTerr56)” School of Biological Sciences, Madurai Kamaraj University. ● Nagakanni M, <u>Soundarya Rani R</u>, Pandima Devi Kasi. Elucidating the apoptotic potential of the synergistic compounds (C1&C2) through <i>in silico</i> and <i>in vitro</i> approach. ● <u>Indhirakumar B</u>, Jafni S, Nagakanni M, Soundarya Rani R, Pandima Devi Kasi. <i>In Silico</i> Evaluation of Flavonoids Against Tau Hyperphosphorylation: Molecular Docking and Pharmacokinetic Perspectives For Alzheimer's Disease.
8th and 9th August, 2025	Harnessing Biotechnology : Driving Innovation in Health Care Agriculture and the Environment (NCBH-25)”, Centre for Drug Discovery and Development, Sathyabama Institute of Science and Technology, Chennai.

	Indhirakumar Balakrishnan, Jafni Sakthivel, Nagakanni Muthuvenugopal , Soundarya Rani Rajendran, Kailash Baskaran, Pandima Devi Kasi. Targeting Tau Hyperphosphorylation in Alzheimer Disease: Molecular Dynamics Analysis of Natural Compounds Against Key Regulatory Proteins.
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NATIONAL LEVEL - POSTER PRESENTATIONS:

27 th - 28 th May, 2022	National Level Conference on Biotechnology and OMICS Sciences, Department of Biotechnology, Alagappa University, Karaikudi, India. Early stage Non-Small Cell Lung Adenocarcinoma: Unveiling Differentially Expressed Genes and Therapeutics using <i>in silico</i> Approaches.
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WORKSHOPS ATTENDED:

1	23 rd - 25 th Jan, 2024	IMMUNOTECH - 2024, Workshop on “Immunohistochemistry and In Vivo Imaging Technologies” Organised by Department of Bioscience and Bioengineering (Cell and Tissue Engineering Lab), IIT Bombay, Maharashtra.
2	17 th - 19 th Jan 2024	“Hands-on Workshop in Flow Cytometry, Cell Sorting, Data Analysis and Advance Applications” at FACS Facility, IIT Bombay, Maharashtra.

CONFERENCES ATTENDED:

1.	23 rd - 25 th Jan, 2023	International Conference on Innovations and Applications in Biomedical and Health Sciences - conducted by PG and Research Department of Biotechnology and Bioinformatics, Holy Cross College, Tiruchirapalli, India in collaboration with Department of Anatomy, Yong Lin School of Medicine, National University of Singapore and Department of Endocrinology, Dr.ALM PG IBMS, University of Madras, Taramani Campus, Chennai.
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2.	27 th - 28 th May, 2022	National Level Conference on Biotechnology and OMICS Sciences BIO-OMICS - 2022, conducted by the Department of Biotechnology, Alagappa University, Karaikudi.
3.	13 th - 14 th May, 2022	Symposium on Biological Diversity - National Perspective, Biodiversity Crisis, Concerns and Mitigation Approaches jointly organized by the Department of Biotechnology and Bhabha Atomic Research Centre.
4.	11 th October 2021	“Mesoporous Nanomaterials in Cancer Therapeutics” webinar organized under MHRD-RUSA 2.0 “Biomaterials” project by Center of Advanced Study in Crystallography and Biophysics, University of Madras
5.	15 th September 2021	“ROR1 TARGETED THERAPY - A BOON FOR METASTATIC CANCER TREATMENT” Organized by the Department of Biochemistry, Saveetha Dental College & Hospital.
6.	25 th - 26 th February, 2021	International Virtual Conference On Drug Discovery And Translational Research Hosted By The Department Of Biochemistry & Molecular Biology, Pondicherry University.
7.	20 th - 24 th July 2020	COVINAR - A week long International webinar on COVID - 19 organized by the department of Biochemistry and Molecular Biology, Pondicherry University.
8.	7 th July, 2020	virtual conference on recent trends in plant genetics and genomics organized by VIT school of Agricultural Innovations and Advanced Learning (VAIAL).
9.	18 th June, 2020	International webinar "Scientific Communication for Postgraduate Students and Young Researchers in Life Sciences" organized by School of Life Sciences, Pondicherry University.
10.	3 rd - 4 th June, 2020	International Online Seminar on “Alternatives to Antibiotics” organized by Department of Biotechnology, Alagappa University sponsored by RUSA 2.0.
11.	20 th April - 2 nd May, 2020	Webinar Series on “Novel Microbes and Newer Threats” organized by Sathyabama institute of science and technology in association with Indian Association of Applied Microbiology(IAAM) California University of Science and Medicine,USA.
12.	12 th - 15 th , 2020	SFRR-INDIA-2020 Conference on “Role and management of oxidative stress in human disease” organized by Bioscience Group, Bhabha Atomic Research Center, Mumbai.

13.	09 th - 13 th September, 2019	International Conference cum workshop on <i>Caenorhabditis elegans</i> based OMICS for future Challenges organized by the Department of Biotechnology, Alagappa University.
14.	9 th - 10 th , 2019	Theme meeting on Environmental Receptivity of Industrial Everything and their influence on Marine Ecology & DAE technologies for Water Treatment Sponsored by Board of Research in New Science & Alagappa University, Jointly Organized by Bhabha Atomic Research Center
15.	6 th - 8 th February, 2019	UGC & DBT Sponsored National Seminar on Genomics and Proteomics organized by the department of Biotechnology.
16.	8 - 9 th January, 2018	Entrepreneurship Awareness Camp Jointly Organized by the Center for Entrepreneurship Development (Tamil Nadu) & Department of Botany, The American College, Madurai - Sponsored by Ministry of Science And Technology (NSTEDB, Government of India, New Delhi).

CO-MENTORING:

1.	V. Manimegalai Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Screening of Novel Phytocompounds for Its Anti-Cancer Property Through <i>In Silico</i> and <i>In Vitro</i> Studies
2.	R. Revathi Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Evaluation of Natural Compounds Targeted Against Non-Small Cell Lung Cancer: <i>In Silico</i> And <i>In Vitro</i> Assessment
3	Shubhangi Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Evaluating the Combination of Citral and Cisplatin for its Anti-cancer Activity Through Apoptotic Pathway: <i>In silico</i> and <i>In vitro</i> Approach
4	Srinithi K Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Sequential Combination of Citral with Cisplatin Targeting Lung Adenocarcinoma Cells Through ROS Pathway.
5	Suruthi M - Mini Project	Evaluation of Anti- Breast Cancerous Effect of

	Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Synergistic Phytocompounds Through <i>In Silico</i> and <i>In Vitro</i> Approaches
6	Sangita Behera Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Evaluation of Anti-Cancer Property of Farnesol Through <i>In Silico</i> And <i>In Vitro</i> 2D And 3D Approaches
7	Padmesh S Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Proteomics analysis of anti-metastatic potential of farnesol in Non-small cell lung cancer cells
8	Srilekha M K Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Apoptotic Potential of Farnesol Against A549 Lung Adenocarcinoma Cells Through Inhibition of Anaerobic Glycolysis – An <i>In Silico</i> and <i>In Vitro</i> Study
9	Vinothini - Mini Project Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Profiling the EMT inhibitory potential of natural bioactive compounds against NSCLC: An approach to inhibit lung metastasis and progression
10	Rani Joseph Graduate Student, Department of Biotechnology, Alagappa University, Karaikudi, India.	Assessment of Anti-Metastatic Potential of Phytocompound (A) Against Lung Adenocarcinoma: An <i>In Silico</i> and <i>In Vitro</i> Approach

ORGANIZATION SKILLS:

- Assisted in conducting the Council of Scientific and Industrial Research (CSIR) supported National Level Conference on Biotechnology and OMICS Sciences held on May 27th & 28th, 2022.

SKILL SETS

<u>Techniques:</u>	Basic Techniques: - Microscopy and UV Spectrophotometry - Handling microbes of Biosafety level II - Handling cell lines
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	• Plant tissue culture Molecular Biology Techniques: • DNA, RNA, and plasmid isolation and purification • real-time qPCR • Gel electrophoresis (agarose, SDS PAGE and NATIVE PAGE) • Gene cloning and transformation • Primer designing • Triparental mating
<u>Software:</u>	• BioRender (Toronto, Canada) for scientific figures. • OriginPro version 8.5 (OriginLab®, USA). • GraphPad Prism Version 10.5.0 Bioinformatic Tools: • PyRx 0.8 • Discovery Studio Visualizer - BIOVIA - Dassault systèmes.

REFERENCE:

Dr. K. Pandima Devi Professor Department of Biotechnology Science Campus Alagappa University Karaikudi - 630003 Contact: + 91 4565 223325 E-mail ID: devikasi@yahoo.com	Dr. S. Karutha Pandian Former Senior professor and Head Department of Biotechnology Science Campus Alagappa University Karaikudi- 630003, Tamil Nadu Contact: + 91- 4565 225215 E-mail ID: sk_pandian@rediffmail.com E-mail ID: pandiansk@gmail.com
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